

JULIANNA RAINE YU LACADEN

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EDUCATION

Saint Louis University Bachelor of Science in Computer Science	Baguio City, Philippines 2023-Present
- Relevant Coursework: Software Engineering, Operating Systems, Database Systems, Artificial Intelligence, Website Development, Application Development. - Dean's Lister for 4 consecutive semesters from AY 2023-2025	
Merryland Montessori and High School Inc.	Urdaneta City, Pangasinan
- Secondary Education (2022-2023) - Elementary Education (2016-2017)	

PROFESSIONAL EXPERIENCE

EcoSolergy Solar Power Engineering Services <i>Sole Software Developer</i>	Urdaneta City, Pangasinan April 2026 – June 2026
- Functioned as the autonomous developer overseeing the end-to-end lifecycle, from requirements gathering to deployment of EcoSolergy, a secure WPF desktop application built on .NET 8. - Designed and architected a "Zero-Install" USB-portable architecture using SQLite to eliminate client deployment friction and enable seamless on-site operations. - Engineered a transactional inventory asset workflow handling real-time stock deductions and multi-state automated restorations under an SQLite WAL journal mode framework to guarantee database integrity. - Programmed a high-performance reporting engine utilizing ClosedXML to programmatically generate complex multi-sheet corporate Excel proposals matching strict client templates. - Implemented a resilient, automated daily database backup engine using SQLite's native 'VACUUM INTO' command, incorporating multi-threaded failure logging to eliminate silent backup dropouts.	

INTERNSHIP

Makerspace Innovhub OPC <i>AI Engineering Intern</i>	Mapandan, Pangasinan June 2026 – July 2026
<i>Python · LangGraph · FastMCP · Google Genkit · Playwright · scikit-learn · Ollama · Gemini · Pydantic · FastAPI/SSE</i>	
- Designed and built a two-stage agentic SEO/AEO/GEO audit skill (deterministic Playwright/BeautifulSoup extraction + rule-based 0–100 scoring) packaged as a portable agent skill for Claude Code, OpenCode, and Kilo Code, validating every score point back to a specific extracted field across live sites including toyota.com.ph (71.5/100) and merriam-webster.com (89.1/100). - Built an end-to-end NLP content-rating classifier (TF-IDF + scikit-learn) on a Disney+ titles dataset, reaching 0.527 macro F1, a 13.2x improvement over the majority-class baseline and then extended it with spam/harm and sentiment (IMDB) classifiers deployed on an interactive Streamlit dashboard. - Diagnosed and resolved an HTTP/SSE transport incompatibility to integrate Google Genkit with an existing FastMCP shopping-assistant server, and documented a local-LLM schema-validation failure (typed parameters emitted as strings) with three proposed remediation paths. - Hand-built a multi-provider (cloud + local) tool-calling research agent in LangGraph/TypeScript with credibility-scored, conditional loop-back retrieval, a diagnosing and fixing a state-reducer bug that caused duplicate re-scoring of sources on loop-back.	

LEADERSHIP EXPERIENCE

SLU Integrated Confederacy <i>Business Manager</i>	2024-2026
- Oversee organizational budgeting and resource allocation for activities and events. - Maintain financial documentation and ensure transparency across all expenditures. - Collaborate with officers to support event planning and member engagement initiatives. - Assist in organizational decision-making and strategic planning.	

HACKATHONS & INNOVATION ACCOMPLISHMENTS

AMD Developer Hackathon (Act II) — "Token Gate," Track 1 <i>Python · Ollama · Fireworks AI · Docker · pytest</i>	July 6-11 2026
- Owned local/cloud model integration for a cost-aware LLM router that escalates from a free local model to paid cloud inference only on low-confidence responses, judged on an accuracy gate then ranked by cloud token efficiency. - Found and fixed an incorrect model-ID bug causing 100% of cloud API calls to silently fail; after the fix, verified local-eval accuracy rose from 75% to 100%. - Implemented task-based query classification, a self-consistency confidence check, and a token-budget guard to keep cloud spend within grading constraints. - Hardened the Docker deployment for 4GB RAM / 2 vCPU / 60s cold-start grading limits, cutting compressed image size ~50% (1.5–1.8GB → 750–900MB); top committer on the team (23 of 58 commits) with a full 36/36 test suite passing.	
RainWatt (Rain-Powered Micro-Hydropower System) <i>Sustainability Researcher & Engineer</i>	Baguio Smart City Challenge 2025
Designed a rain-powered micro-hydropower system converting water flow into stored electricity.	

- Engineered turbine-based generation using three 80V water turbines routed through sediment filtration.
- Documented system flow from gutter collection, filtration, turbines, LiFePo battery, household power integration.
- Contributed to sustainability research focusing on renewable microgeneration and water recycling.

TECHNICAL SKILLS & PROJECTS

AI/Agentic: LangGraph, Google Genkit, FastMCP/MCP, Ollama, Gemini, Fireworks AI, Pydantic, scikit-learn, TF-IDF, Streamlit, Playwright, BeautifulSoup4

Languages: C#, Java, Python, JavaScript, TypeScript, PHP, Rust, C++, SQL

Frameworks & Libraries: FastAPI, WPF/.NET 8, Node.js, Docker, pytest, Git/GitHub, ClosedXML, MigraDoc/PdfSharp

Databases & Hardware: SQLite, MySQL, phpMyAdmin, Arduino, LoRa, GPS/IMU sensors

Paybach: Bidding and Swapping Marketplace – Team Project Nov. 2025 – Dec. 2025

- Designed and built a distributed bidding and item-swapping web marketplace optimized for a campus environment using a hybrid PHP and Node.js backend.
- Formulated custom stateful bidding logic, secure product listing pipelines, and data-driven analytical dashboards for user tracking.
- Deployed, configured, and hardened the full-stack system on a secure Ubuntu Server instance, managing database integrity and network traffic optimization

Laboratory Reservation System – Academic Project Nov. 2025 – Dec. 2025

- Developed a robust automated reservation platform using HTML, CSS, JavaScript, PHP, and phpMyAdmin to streamline resource scheduling across university computer laboratories.
- Implemented role-based access control (RBAC) mechanisms to manage multi-tier account privileges, real-time schedule synchronization, and comprehensive administrative override controls.
- Designed a responsive UI focusing strictly on ease of navigation, accessibility compliance, and low-latency rendering.

SOFT SKILLS

- Problem Solving
 - Critical Thinking
 - Collaboration & Communication
- Research
 - Project Management
 - Attention to Detail

LICENSES

Civil Service Eligibility

CSC Regional Office CAR

August 2025

- Career Service Examination: Pen and Paper Test - Professional